Lab 2 Report: EDA and Conversion Funnel Analysis using Python

Course: Marketing Analytics (DS 4115)
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GitHub: https://github.com/MUSHFIQUR-RAHMAN12/Marketing_Analytics

1 Objective

In this lab I moved from Excel to Python. I generated a fake GA4-style session log, built a three step conversion funnel (session start, add to cart, purchase), cleaned the missing values with pandas, and then used a seaborn heatmap to see which variables are linked to purchases. The code is in the notebook `Lab2_Conversion_Funnel_EDA.ipynb`, which is uploaded to my GitHub repository: https://github.com/MUSHFIQUR-RAHMAN12/Marketing_Analytics.

2 Data

The data was generated inside the notebook with `np.random.seed(42)`, so the numbers can be reproduced. It has 5,000 sessions and 8 columns: session ID, user ID, device category, the three funnel flags, page views and time on site (in seconds). About 50 values of `time_on_site_sec` were deleted on purpose so that they could be cleaned later. `df.info()` confirmed this: that column had 4,950 non-null values out of 5,000 and every other column was complete.

3 Conversion funnel

Out of 5,000 sessions, 1,470 added something to the cart and 905 ended in a purchase.

Table 1: Funnel summary

Step	Sessions	Rate vs. step 1 (%)	Drop-off from previous (%)
1. Session Start	5,000	100.0	–
2. Add to Cart	1,470	29.4	70.6
3. Purchase	905	18.1	38.4

The biggest loss is at the first step: about 71% of sessions never add anything to the cart. Of the people who do add something, about 62% buy it, so the checkout step is comparatively healthy. If I had to improve one thing, I would work on getting visitors to add to cart (better product pages, clearer prices, etc.) before touching the checkout.

4 Data cleaning

Only `time_on_site_sec` had missing values (50 of them). I filled them with the median, which is 144 seconds. I used the median instead of the mean because time on site can have a few very long sessions

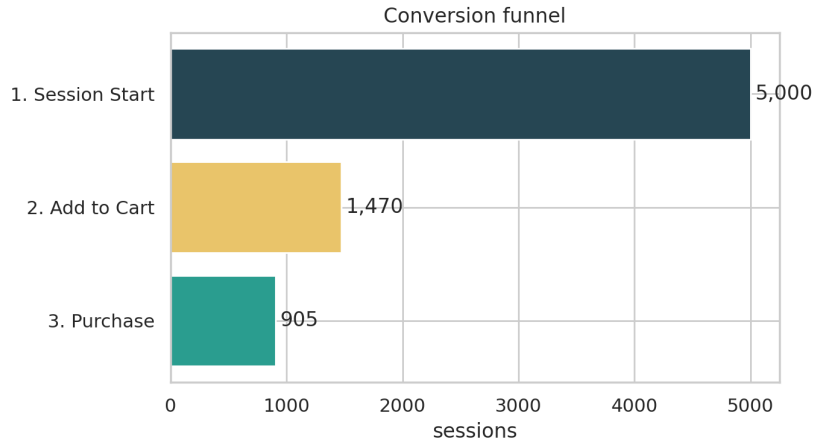


Figure 1: Conversion funnel

which would pull the mean up. I also dropped rows with a missing session ID (there were none) and made the device names lowercase and stripped extra spaces. After this there were 0 missing values left.

5 EDA and correlation heatmap

Conversion by device was very similar: the add to cart rate was between 28.9% and 30.3% and the purchase rate between 17.6% and 18.6% for desktop, mobile and tablet, so device does not look like a blocker here.

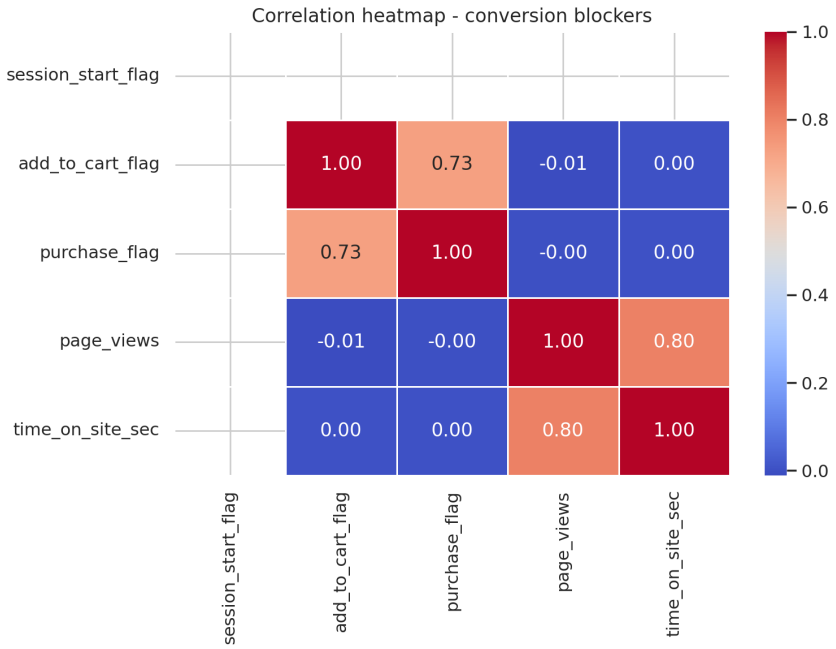


Figure 2: Correlation heatmap

What I noticed in the heatmap:

- **Add to cart and purchase** have a correlation of 0.73. This is expected, because in the data only people with a cart can buy.

- **Page views and time on site** have a correlation of 0.80, since time on site is calculated from page views.
- **Page views and time with purchase** are almost 0 (between -0.01 and 0.004). So in this dataset, spending more time or viewing more pages does not make a purchase more likely.
- The **session start** row and column are empty. That flag is 1 for every session, so it has no variance and the correlation cannot be calculated.

6 Conclusion

The funnel shows that the main conversion blocker is getting people to add to cart, with 70.6% leaving at that step. Engagement measures (page views and time) were not related to buying, but I should say that this is because the dataset is random. With real GA4 data I would expect engagement to matter, so the conclusion about those two variables should not be taken to a real business. The main thing I learned is how to build a funnel and clean data in pandas, and to check what a heatmap is really showing before interpreting it.